Neurolabs iOS SDK Integration Guide

Technical Integration Specification

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1. Scope

This guide is for partner iOS apps integrating ‘NeurolabsSDK‘ through Swift Package Manager.

2. Release History

No public API contract has been broken since v1.1.9. New options have been added with backwards-compatible defaults. Recommended defaults: strict shelf guidance with overlays disabled, ‘liveQualityEnabled: true‘, ‘autoCloseAfterCapture: true‘ for single-capture parity flows. ‘NLCameraConfiguration.captureRange(...)‘ is the preferred helper for custom-camera flows that need a minimum capture count and a higher maximum cap. ‘showsCapturedCountLabel‘ hides the ‘N captured‘ chip in custom camera and multi-capture flow UIs.

v1.7.29

No initialiser changes and no removals. New APIs are additive.

New APIs: ‘NLCaptureMode.puzzle‘ (section 9); ‘NLMultiBayResult.layout‘ (‘NLMultiBayLayout‘) and ‘.tiles‘ (‘NLMultiBayTile‘, ‘NLPlaneRect‘) and ‘NLMergedProduct.sourceDetectionIds‘ (section 9); ‘layout‘ and ‘tiles‘ on the bridge’s ‘NLNativeMultiBayResultSummary‘ (‘NLNativeMultiBayTile‘); ‘NLCameraConfiguration.arProductMarkers‘ with ‘NLARProductMarkerStyle‘ (section 13); and the ‘NLTheme‘ presets ‘NLTheme.ziaLens‘ (the default) and ‘NLTheme.neutral‘ (the opt-out), its tokens, font providers and ‘NLTypeRole‘ (section 14, "Theming the SDK screens"). A host that switches over ‘NLCaptureMode‘ without ‘@unknown default‘ needs the ‘.puzzle‘ case.

Behaviour changes, with no opt-in:

- The default look is the ZIA Lens v3 design system: ‘NLTheme.current‘ and the AR capture camera’s chrome change colour, shape and glyphs. To keep the previous look, set ‘NLTheme.current = .neutral‘ once at setup (section 14). The bridge has no theme field, so a WebView or Cordova app opts out in its native layer.
- The AR screen takes the recognition lab’s guidance chrome: the instruction sits in the middle of the screen, a level marker shows square and level, and the copy reads "Step left" / "Step right" and "Capture".

v1.7.28

No initialiser changes and no removals. New APIs are additive.

New APIs: ‘NLValidationIssueKind.shelfFramingPoor‘ and the shelf-framing properties on ‘NLValidationOptions‘ (section 7, "Shelf framing and Retake recommended"); the fixture API v2 methods on ‘client.outlets‘ and ‘client.missions‘, with ‘NLFixtureAPIError‘ (section 5); the testing seam ‘NLFrameSource‘ (with ‘NLFrameSourcePhoto‘) / ‘NLDetectionSource‘ through ‘NLCameraConfiguration.frameSource‘ and ‘.detectionSource‘, ‘isSynthetic‘ on captures and the error code ‘SYNTHETIC_CAPTURE_REFUSED‘ (section 6); ‘NLJoinDegradedReason.snapshotMismatch‘ (section 12); and ‘NLOperationsHttpError.retryAfter‘ in ‘OperationsCore‘. ‘Outlets.addOutletSubsections‘ is deprecated; use ‘addOutletFixture‘. A host that switches over ‘NLValidationIssueKind‘ or ‘NLJoinDegradedReason‘ without ‘@unknown default‘ needs the new cases.

Behaviour changes, with no opt-in:

- A still whose detections show a badly framed shelf carries a ‘shelfFramingPoor‘ issue (severity ERROR) but stays acceptable, and the capture review reads a red "Retake recommended" for a kept photo with an ERROR (section 7). Turn the check off with ‘enableShelfFramingCheck = false‘.
- During scheduled maintenance (a 503 with ‘error_code: maintenance‘) a queued photo stays pending with ‘lastErrorCode‘ ‘UPLOAD_MAINTENANCE‘ and spends no retry, for up to 24 h (section 6).
- Operations client calls wait out a 503’s ‘Retry-After‘ (capped at 60 s) when they retry (section 5).
- With ‘mirrorGate‘ set, a snapshot granted under a different id from the one the config announced is gated too (section 12).

v1.7.27

No initialiser changes, no new APIs and no removals.

Behaviour changes, with no opt-in:

- A guided (shutter) multi-bay walk, and any multi-bay session on the default camera, may plan up to 20 bays: ‘NLMultiBayOptions.maxSupportedBays‘ is 20 (it was 8), and ‘init‘ clamps ‘maxBays‘ to it. The default stays 8 bays. A host that asked for 9-20 bays and was clamped to 8 now gets what it asked for (section 9). A continuous AR sweep keeps its 30.

v1.7.26

No initialiser changes and no removals. New APIs are additive.

New APIs: in ‘RecognitionBootstrap‘, ‘NLRecognitionBootstrap.Configuration.mirrorGate‘ (a property, not an ‘init‘ parameter), ‘NLRecognitionBootstrap.joinDegraded‘ and ‘.onJoinDegradedChange‘; in ‘RecognitionInterface‘, the gate’s types (‘NLMirrorGateConfiguration‘, ‘NLCatalogMirrorPositionProvider‘, ‘NLMirrorStatus‘, ‘NLMirrorGatePolicy‘, ‘NLJoinDegradedReason‘ and the rest) and ‘NLSnapshotSyncManager.enableMirrorGate(provider:policy:store:)‘ / ‘syncGated(publishedAt:now:catalogPosition:)‘ (section 12, "Catalogue-mirror gate"). ‘NLDetectorModels.statusDidChangeNotification‘ carries ‘userInfo["reason"]‘ (section 7).

Behaviour changes, with no opt-in:

- ‘stopSession(_:)‘ waits, at most 5 s, for the session’s ‘enqueueCapture‘ calls already running (section 10, "Queue before you stop the session").

- A core with no API key and no auth provider starts no detector download, and a ‘no_credentials‘ raised before any request sets no retry wait, so a credential set right after the core’s init downloads the detector.
- The detector chip and the AR dock name the fix for ‘insufficient_storage‘ (free about 200 MB) and ‘waiting_for_unmetered_network‘ (connect to Wi-Fi). The AR walk shows one detector message at a time and waits 45 s for the model at most once per process.
- A long-lived operations client binds the credential when it is built: read ‘sdk.operationsClient‘ again after a key rotation (section 5).

v1.7.25

No initialiser changes and no removals. New APIs are additive.

New APIs: in ‘NeurolabsSDK‘, ‘NeurolabsSDKCore.computeVisitKPIs(forSession:context:storeContext:)‘, with the new error code ‘VISIT_KPIS_SESSION_UNKNOWN‘ (section 14, "Visit KPIs by session id"), and ‘NeurolabsSDKCore.defaultProductNamesEnabled‘ (default ‘true‘) for the AR live labels’ default names (section 13). A queued upload can report the new retryable code ‘UPLOAD_CREDENTIALS_CHANGED‘.

Behaviour changes, with no opt-in:

- ‘autoCloseAfterCapture‘ on an embedded camera closes only at the capture target, or on Done when there is a Done button, and emits ‘onFinish‘ with the session’s photos. It used to close after the first photo with no ‘onFinish‘. A session ended with lists the photos saved from the preview (‘cancelled: true‘); they were queued at their Save. With auto-close, queue before you stop the session (section 10).
- Multi-bay walks deduplicate the products two bays share, as on Android: counts, merged products and ‘device_dedup‘ drop to the deduplicated count, and a pair of bays that cannot be lined up splits the walk (section 9).
- A crop forgets the recognition of the photo as shot, and a photo that had a pass is recognised again on the crop (section 10, "Photo cropped").
- ‘priceReadings(forCapture:)[].pairedDetectionId‘ pairs with the same shelf geometry as K16 (section 14).
- An operations retry never sends a rotated credential to the old host: a request whose base, key or auth provider changed after its first attempt is aborted.

v1.7.24

No initialiser changes and no removals. New APIs are additive, and the AR ones are experimental and off by default.

New APIs: in ‘NeurolabsSDK‘, ‘NLMultiBayCountSource‘, ‘NLCameraConfiguration.multiBayCountSource‘ and ‘.arLiveKPIRings‘ (the same keys on ‘NLNativeCaptureConfig‘ and ‘NLNativeCaptureOptions‘), ‘NLMultiBayResult.countSource‘, ‘.aggregatorCountsByLabel‘, ‘.aggregatorProductCount‘ and ‘.perBayCountsByLabel‘, ‘NLNativeMultiBayResultSummary.countSource‘ and ‘.aggregatorCountsByLabel‘ (section 9 and section 13), ‘NLVisitKPIHostStoreContext‘ and ‘NeurolabsSDKCore.computeVisitKPIs(for:context:storeContext:)‘ (section 14).

Behaviour changes, with no opt-in:

- Live camera frames reach the detector upright, so the AR camera detects products live again: live boxes, the live chip and the live guidance work (section 7). Photos were never affected. Frame stats (glare, a finger over a corner) are read upright too.
- Long AR walks stay inside the track budget: out-of-view facings are archived and still count, so the last bays get boxes. Under ‘.v1‘ the multi-bay AR count (‘ar_anchor_counts‘) counts archived facings too (section 13).
- The AR multi-bay walk waits up to 45 s for the detector before it captures. The review says "No catalogue matches" when nothing matched, and a remembered Guided / Auto pick applies only under the host default it was made under (section 13).
- The detection inspector names a match only at 0.5 or more (section 12).
- K16 price-tag coverage pairs tags by shelf geometry, so ‘tag_paired‘ no longer reads about 0 % on full-resolution photos (section 14).
- Visit KPIs at ‘stopSession(_:)‘ survive an app kill: every photo counts, with its recognition matches (section 14, "Visit KPIs").
- K9-K11 and N4 / N5 are pinned to the shared reference rows; the values are unchanged.

v1.7.23

No public API changes: no new APIs, no initialiser changes and no removals.

Behaviour changes, with no opt-in:

- Snapshot manifests must be signed. The SDK pins the backend’s snapshot keys (the staging key only on a Neurolabs staging operations host), and a manifest that is unsigned, signed by another key or altered fails to load, so recognition stays off for that build (section 12). Every live manifest has been signed since 18 Oct 2026, and older SDKs ignore the signature. A snapshot an older SDK cached for offline use is unsigned and is not used offline; the next online load replaces it. An ‘NLCatalogSnapshotLoader‘ you build with the default ‘pinnedPublicKey‘ now refuses unsigned manifests.
- With your own capture UI (‘NLCustomCameraView‘, or photos you queue with ‘enqueueCapture‘), ‘stopSession(_:)‘ computes the visit’s KPIs once and ‘device_kpis‘ rides the submit under the usual rules. A visit that spans an app kill gets a partial count (section 14, "Visit KPIs"). Set the context on the session you stop.
- ‘quality.capture_verdicts‘ counts a capture as ‘warn‘ when its quality check judged it not acceptable or flagged a warning or an error.
- The AR count v2 runs in shadow behind the internal ‘.v2‘ fusion profile, which is off; ‘.v1‘ is unchanged.

v1.7.22

No initialiser changes and no removals. New APIs are additive, with one source change: ‘NLCaptureDeletionSource‘ gains ‘.retake‘, so a ‘switch‘ over it without ‘@unknown default‘ must handle the new case.

If your app keeps photos from ‘onCapture‘ (save handlers that return ‘.skipDefault‘):

- A retake now reaches ‘onCaptureDeleted‘ with source ‘.retake‘: drop that photo. A crop reaches the new ‘NLCameraConfiguration.onCaptureCropReplaced‘ (‘NeurolabsSDKCore.onCaptureCropReplaced‘ for ‘openNativeCaptureUI‘) with the cropped photo: swap your copy of ‘replacedCaptureId‘. Both are experimental (section 10).
- With a recognition provider set, the camera recognises those photos before its save flow returns (at most 10 s), so ‘recognitionMatches(forCapture:)‘ and ‘NLCapturePhoto.recognitionMatches‘ are filled. Nothing new is queued or uploaded. With an embedded camera, set ‘NeurolabsSDKCore.shared‘ to the core your save handlers commit through if your app creates more than one core (section 12). ‘recognizeCapture(captureId:imageData:detections:)‘ recognises a photo outside the camera (section 12).

Behaviour changes, with no opt-in:

- "Face the shelf straight on" (the shelf-row yaw leg) fires above 12 and clears at 9 or less (was 8 and 5). Every org’s ‘config_hash‘ in ‘guidance_config_applied‘ changes with this release.
- With ‘uploadsEnabled == false‘, a credential no longer opens the SDK’s background ‘URLSession‘, fetches ‘GET /v1/config‘ or posts an owed submit.
- A ‘NeurolabsSDKCore‘ created without a key or auth provider keeps the detector credential you registered with ‘NLDetectorModels.setCredentials‘.
- Releasing an ‘NLRecognitionBootstrap‘ withdraws its embedder from the AR camera’s appearance step: keep the bootstrap while you want it used.
- The AR live "~ N products" chip counts SKU facings only, and shutter presses during limited ARKit tracking are not fused. Served counts are unchanged.

New APIs: in ‘NeurolabsSDK‘, ‘NLCaptureDeletionSource.retake‘, ‘NLCaptureCropReplacement‘, ‘NLCameraConfiguration.onCaptureCropReplaced‘, ‘NeurolabsSDKCore.onCaptureCropReplaced‘, ‘NeurolabsSDKCore.recognizeCapture(captureId:imageData:detections:)‘, ‘NeurolabsSDKCore.arSessionRecording‘ (experimental, section 7) and ‘NLSDKDebug.guidanceDiagnostics‘; in ‘RecognitionInterface‘, ‘NLOnDeviceEmbedderRendezvous.publishWithdrawable(_:)‘, ‘withdraw(_:)‘, ‘subscribeToWithdrawals(_:)‘, ‘Publication‘, ‘Embedder.publicationId‘ and ‘didWithdrawNotification‘.

v1.7.21

No initialiser changes. New APIs are additive. One breaking removal: ‘NLShelfDetector‘ and ‘NLCoreMLYoloDetector‘ are gone. The SDK never constructed them; if your app called them directly, use the SDK’s built-in detection instead.

Behaviour changes, with no opt-in:

- An embedded camera (‘NLCustomCameraView‘, ‘NLCustomCameraViewController‘) that leaves the screen (a push, a tab switch, a modal over it) pauses and resumes on return; its staged photos are kept. The session ends on close, Done or ‘closeNativeCaptureUI‘. An embedded controller must pass ‘onClose‘.
- Opening the SDK camera right after a close waits for the dismissal. An open the SDK cannot present reports ‘didEncounterError‘ with the new code ‘CAMERA_PRESENT_FAILED‘, and a camera that cannot start reports ‘CAMERA_UNAVAILABLE‘. Neither is sent to Sentry.

- In strict mode the camera drops a failed photo before your ‘onCapture‘ hears of it. ‘NLNativeCaptureResult.validationResult‘ carries the quality verdict (nil when nothing validated the capture).
- The SDK camera (‘openNativeCaptureUI‘) refuses a photo before the shutter when the queue would refuse it (‘STORAGE_LOW‘, ‘QUEUE_FULL‘); you get ‘didEncounterError‘ with no capture result.
- Image quality reads luma as Rec.601, as on Android: a saturated colour can get a different verdict than on 1.7.20. No threshold changed.

New APIs (section 14):

- Store context: ‘prefetchStoreContext(outletIds:)‘, ‘storeContextStatus(outletId:)‘, ‘setStoreContextIdentity(_:)‘, ‘clearStoreContextCache()‘.
- Visit KPIs: ‘neurolabsSDK(_:didComputeVisitKPIs:sessionId:)‘ (default no-op), ‘setVisitKPIsEnabled(_:)‘ (on by default), ‘setVisitKPIContext(_:forSession:)‘, ‘computeVisitKPIs(for:context:)‘.

v1.7.20

No API changes. It ships the REX/Smollan audit fixes; 1.7.19 is unchanged, so upgrade to 1.7.20 to get them.

Behaviour changes, with no opt-in:

- With Wi-Fi-only uploads on, a transfer iOS finishes in the background never uses cellular data. The SDK hands a photo to iOS only while the app is in the foreground, and on the next foreground restarts a transfer that 1.7.19 handed off in the background and that has sent nothing (section 6).
- ‘retryFailedUploads()‘ leaves a mission refusal that cannot succeed as sent failed (only ‘flushQueue(force: true)‘ re-sends it), and a call within 60 s of the previous one does nothing (section 6).
- The SDK’s default camera tells the rep why it refused a photo for low storage or a full queue; the host signal is unchanged (section 6).
- "Preparing detector..." holds the shutter for 3 s at most.
- SDK error reports to Sentry never carry a server response body, only the HTTP status and the server’s ‘error_code‘ (section 14).

v1.7.19

No initialiser changes. One additive static method and one stored property set after ‘init‘, both additive for the ABI.

Behaviour changes, with no opt-in:

- A queued photo’s bytes go up through an iOS background ‘URLSession‘, so they keep uploading after the app is suspended or terminated by iOS, and while the phone is locked (not after a user force-quit; see section 6). Forward the app-delegate wake to the SDK to finish sooner (section 6, "Uploads after the app is backgrounded").
- "Review shelf" offers "Retake bay N" on a broken bay link, and an AR multi-bay walk that folds back across a broken link is no longer double-counted (a walk without AR positions makes no attempt).
- The offline recognition build expires after the org’s configured number of days (7 by default) instead of a fixed 7 days.

- The SDK’s process-global Sentry client (started by the Cordova plugin) reports only the SDK’s own errors and crashes, and never replaces a Sentry client your app started first (section 14).

New APIs:

- ‘NeurolabsSDKCore.handleEventsForBackgroundURLSession(_:completionHandler:)‘, optional (section 6).
- ‘NLCameraConfiguration.analyticsEnabled‘ (‘Bool?‘, ‘nil‘ by default: follow the core), so a camera-only app can opt in to camera analytics (section 7, "Camera analytics setting").

v1.7.18

No initialiser changes. The new options and result fields are stored properties and are additive for the ABI.

Behaviour changes, with no opt-in:

- After each photo uploads, its capture metadata (capture contract v1) goes up from a retrying outbox; location and depth only when you opt in (section 6).
- An upload in flight when the app is backgrounded gets iOS’s background grace period (section 6).
- A multi-bay walk with a broken bay link is deduped per linked segment instead of giving up on the rest of the walk; the submit sends ‘device_dedup‘ v2 when every segment is trusted, and the walk’s stitched realogram (section 9).
- With on-device recognition off, the AR overlay tells touching boxes apart by appearance (section 13).
- The camera follows the core’s ‘analyticsEnabled‘, and a camera opened before any ‘NeurolabsSDKCore‘ exists sends no analytics (section 7, "Camera analytics").

New APIs:

- ‘NLConfiguration.captureGeolocation‘ and ‘NLConfiguration.captureDepth‘, both off by default (section 6).
- ‘NLNativeCaptureResult.captureMetadata‘ and ‘NLCustomCameraCapture.captureMetadata‘ (‘NLCaptureMetadata‘, section 6).
- ‘NLMultiBayResult.segments‘ (‘[NLDedupSegment]‘) and ‘NLMultiBayResult.realogram‘ (‘NLStitchedRealogram‘), also on ‘NLNativeMultiBayResultSummary‘ for bridge hosts (section 9).
- ‘NLCameraConfiguration.arAppearanceWithoutRecognition‘, on by default, and its bridge key on ‘NLNativeCaptureConfig‘ / ‘NLNativeCaptureOptions‘ (sections 9 and 13).

v1.7.17

No initialiser changes. The new options are stored properties set after ‘init‘ and are additive for the ABI.

Behaviour changes, with no opt-in:

- An offline chip on the default and AR camera, on by default (section 7).
- "Review shelf" pages through a sweep of more than 8 bays and lists the walk’s brands and products under "On this shelf" (section 9).

- The bridge (‘NLNativeMultiBayConfig‘, WebView and Cordova) carries a continuous AR sweep’s plan of up to 30 bays (section 9).

New APIs:

- ‘NLCameraConfiguration.showsOfflineIndicator‘ (section 7).
- ‘NLCameraConfiguration.arBoxUnmatchedColor‘ (section 13).
- ‘NeurolabsSDKCore.productDetailsSource‘: product names for a camera opened by ‘openNativeCaptureUI‘ (section 13).
- The bridge keys ‘showsOfflineIndicator‘ and ‘arBoxUnmatchedColor‘ on ‘NLNativeCaptureConfig‘ and ‘NLNativeCaptureOptions‘.

v1.7.16

No initialiser changes. New per-camera options are stored properties set after ‘init‘ and default to the old behaviour; the retained-photo settings and ‘prefetchPolicy‘ are process-wide statics.

Behaviour changes, with no opt-in:

- A new capture is refused with ‘STORAGE_LOW‘ when the queue’s directory has under 500 MB free (section 6). There is no switch to turn this off.
- Captures are sized to a 4032 px long edge at JPEG 0.9 when the host sets no ‘maxImageDimension‘ / ‘imageCompressionQuality‘, and the org’s remote ‘sdkCapture‘ values can resize them further for hosts that set nothing (section 7).
- The AR shutter queues ARKit’s 12 MP still instead of the ~1920x1440 video frame (section 13).

New APIs:

- ‘finalReview‘: an end-of-session review on the default camera (section 7).
- Retained photos: ‘NLConfiguration.maxRetainedPhotos‘, ‘retainedPhotoExpiryDays‘, the ‘retainedPhotosHigh‘ delegate methods, ‘STORAGE_LOW‘, ‘NLQueueStatus.retainedCount‘ / ‘retainedBytes‘ (section 6).
- ‘NLDetectorModels.prefetchPolicy‘ and the image-only quality checks while there is no detector (section 7).
- Capture images default to 4032 px / JPEG 0.9, overridable per org by the remote config (section 7).
- AR options: ‘arBoxStyle‘, ‘arBoxHighlightColor‘, ‘arCoverageStyle‘, ‘arCoverageOutlineColor‘, ‘arBoxPersistence‘, ‘arProductMarkers‘, ‘arLiveLabels‘, ‘arLiveCount‘, ‘productDetailsSource‘; 12 MP AR stills (section 13); ‘NLMultiBayOptions.maxSupportedContinuousBays‘ (section 9).

v1.3.2

- Mixpanel ingestion routes to the EU region via ‘NLAnalyticsContext.resolveMixpanelEndpoint(fallback: .mixpanelEndpointEU)‘ at SDK construction; the ‘NLMixpanelAnalyticsTracker‘ default itself stays on the standard US endpoint so partners reusing the class for their own US-region projects are unaffected. Endpoint can be overridden via the ‘NLMixpanelEndpoint‘ Info.plist key or the ‘NL_MIXPANEL_ENDPOINT‘ environment variable (env wins over plist).
- Mixpanel ‘time‘ field is now serialized as a JSON number (epoch seconds). Previously ‘time‘ was sent as a string, which Mixpanel rejected with ‘{"status":0,"error":"time field must be a number"}‘, silently dropping every event.

- ‘NLCustomCameraView.saveAllCaptures‘ keeps the Done button responsive: the synchronous ‘FileManager.removeItem‘ loop for staged-capture files moved into a ‘Task.detached(priority: .utility)‘ block, and the Done label swaps to an inline ‘ProgressView‘ while ‘isPersistingCaptures == true‘. Opacity remains at 1 during persist so the indicator stays visible.
- New ‘NLOperationsCatalogClient‘ (in ‘ProductAuditKit‘) for the operations product catalog, with ‘BarcodeLookupIdentity‘ normalising UPC-E/EAN-8/UPC-A/EAN-13 to GTIN-14 end-to-end. ‘NLOperationsCatalogError.invalidBarcode‘ / ‘.invalidSearchQuery‘ / ‘.malformedResponse‘ are thrown rather than crashing the host with ‘precondition‘.
- ‘NLBDemoApp‘ gains an in-app settings sheet on ‘CustomCameraDemoView‘ (gear icon, top-right). Sliders / steppers / toggles for every tunable knob, bindings wired to existing ‘@AppStorage‘ properties so saved values survive across launches.
- Mixpanel event identity hardened: per-install distinct id persisted in ‘UserDefaults‘, ‘$insert_id‘ (UUID4) on every event for server-side dedup, response body surfaced when ‘debugLogging‘ is true.

v1.3.1

- Restrict barcode symbologies to retail formats (EAN-13, EAN-8, UPC-A, UPC-E) so the scanner stops emitting non-product codes during capture.

v1.3.0

- Angle-off warning threshold relaxed to 12 for less aggressive blocking during normal in-store motion.
- Multi-capture preview deletion: persisted thumbnails are removed when the matching capture is dropped.
- AVFoundation preview recovery hardened against backgrounding / interruption.

v1.2.8

- ‘ProductAuditKit‘ test targets added (no public API change).

v1.2.6 / v1.2.7

- Product Audit (onboarding) MVP: serialise ‘AVAssetWriter‘ finalize and ‘OCRSmartRotationProvider‘ work via in-flight ‘Task‘, off-actor keyframe JPEG writes, completedStepIDs reflect steps that actually ran, step transitions serialised via pending-transition queue.

v1.2.5

- New ‘showsSequenceCounterLabel‘ option in the native capture API for toggling the on-screen sequence counter.
- Persistent custom-camera tuning in the demo app (precursor to the v1.3.2 settings sheet) - preferences survive across runs via ‘UserDefaults‘.
- AVFoundation preview recovery improved on capture failure.
- Strict-rejection preview freeze regression fixed.

v1.2.4

- Release build / Swift 6 archive blockers resolved.

v1.2.3

- ‘NLCameraConfiguration.captureRange(...)‘ helper for bounded multi-capture flows.

- 0.5/1 lens switcher built on AVFoundation, default camera mode starts on the ultra-wide lens when available. Tap-to-focus, lens switch fallback covered by unit tests.
- Custom-camera startup, lens switching, and guidance responsiveness improvements; camera preview shown during warmup, capture button gated until ready.
- Default task routing config uses ‘taskUUID‘.
- Recommended custom-camera parity flow remains strict shelf guidance with overlays disabled, ‘liveQualityEnabled: true‘, ‘autoCloseAfterCapture: true‘ with a single required capture.

v1.2.2

- Analytics tracking + opt-out, partner/device context, Mixpanel defaults, Sentry integration with privacy-safe defaults.

v1.2.1

- Initial lens switcher, memory fixes, bottom-bar centering and UI polish.

v1.2.0

- Docs aligned to v1.1.9 requirements baseline; no SDK code change.

v1.1.9 (baseline)

- Init and per-session routing use ‘taskUUID‘.
- Native detector/model warmup enforced in ‘init‘ and guarded in ‘openCamera‘.
- ‘autoCloseAfterCapture: true‘ keeps preview enabled and closes after Save confirmation.

3. Requirements

- iOS 17.0+ (SDK package declares ‘.iOS(.v17)‘)
- Xcode 16.4+ (release toolchain baseline)
- Swift 5.9+ (Swift 6 toolchains supported)

4. Install (SPM)

Add package dependency (mobile-dist or direct distribution manifest) and link ‘NeurolabsSDK‘.

5. SDK Initialization + Warmup

Uploads use the **operations platform** with a **single key**: the ‘apiKey‘ passed to ‘init‘ seeds the operations credential automatically, and every capture routes through the resumable operations mission flow (‘init-resumable chunked PUT complete‘ per image, ‘submit‘ per session) against ‘api.operations.neurolabs.ai‘. No base URL or separate credential call is needed; use ‘setOperationsApiKey(_:baseURL:)‘ only to rotate the key or to point at a non-production host (staging/dev).

```
import NeurolabsSDK

final class CaptureCoordinator {
    let sdk = NeurolabsSDKCore(configuration: .init(
        apiKey: "<OPERATIONS_API_KEY>",
        debugLogging: true,
        taskUUID: "<DEFAULT_TASK_UUID>" // optional IR-task override (never a mission id)
    ))

    func prepare() {
        // SDK starts image pipeline preparation during init.
        // Optional: observe loading progress through delegate.
    }
}
```

After a key rotation (v1.7.26), read ‘sdk.operationsClient‘ again. Each long-lived operations client the core builds binds the credential (base, key or auth provider) when it is built, and checks every request against it before anything is sent. A client kept from before ‘setOperationsApiKey(_:baseURL:)‘ or ‘setOperationsAuthProvider(_:)‘ fails every request with a retryable ‘NLOperationsError.transport‘ ("operations credentials changed during the request") and never sends the new key to the old base. ‘sdk.operationsClient‘ is rebuilt on the next read.

Fixture API v2 (v1.7.28). ‘client.outlets‘ gains ‘listOutletFixtures‘, ‘addOutletFixture‘, ‘getFixture‘, ‘updateFixtureLocationNote‘, ‘retireFixture‘, ‘reactivateFixture‘, ‘replaceFixture‘, ‘fixtureHistory‘, ‘outletFixtureHistory‘, ‘listFixtureTypes‘ and ‘listFixturesUpdatedSince‘; ‘client.missions‘ gains ‘missionChecklist‘. They return untyped ‘NLJSON‘.

- A fixture POST sends an ‘Idempotency-Key‘. Pass your own UUID to reuse it across your own retries; without one the SDK generates one per call and reuses it across its internal retries only.
- These methods throw ‘NLFixtureAPIError‘ for an HTTP or local validation failure (‘status‘, ‘errorCode‘, ‘currentVersion‘, ‘retryAfter‘, ‘message‘), and ‘NLOperationsError‘ for transport and decoding failures. Catch both: a v2 401 is an ‘NLFixtureAPIError‘, so a global ‘catch NLOperationsError.unauthorized‘ handler does not see it.
- ‘Outlets.addOutletSubsections‘ is deprecated in favour of ‘addOutletFixture‘ (unchanged, not removed).

‘Retry-After‘ on a 503 (v1.7.28). When an operations client call retries a 503 that carries ‘Retry-After: <seconds>‘, it waits ‘max(backoff, min(Retry-After, 60 s))‘. Two retries can therefore hold one call about 120 s; cancel the calling task to stop it (it throws ‘CancellationError‘). A 429 keeps waiting the header’s seconds, uncapped.

6. Queue Management

```
Task {
    await sdk.setAutoSyncEnabled(true)
    await sdk.setWifiOnlyUploadsEnabled(false)
    await sdk.setDeletePhotosOnUploadSuccess(true)
    await sdk.setMaxQueueSize(200)
    await sdk.setUploadRetryCount(5)

    let status = await sdk.getQueueStatus()
    print("pending=\(status.pendingCount) failed=\(status.failedCount)")

    await sdk.flushQueue()
    await sdk.retryFailedUploads()
}
```

‘retryFailedUploads()‘ gives failed photos a fresh cycle, except a mission refusal that cannot succeed as sent (‘wrong_status‘, ‘submission_mismatch‘, ‘mission_not_found‘, ‘invalid_subsection_name‘, ‘invalid_subsection_id‘, or any ‘not_claimed_by_*‘); only ‘flushQueue(force: true)‘ re-sends those. Since 1.7.20 a call within 60 s of the previous one does nothing, so a retry button or timer cannot reset every backoff.

Scheduled maintenance (v1.7.28). While zia-operations is in maintenance its writes answer ‘503‘ with ‘error_code: maintenance‘. A queued photo that gets this answer stays ‘.pending‘ with ‘lastErrorCode == "UPLOAD_MAINTENANCE"‘, spends no retry, and is tried again after the server’s ‘Retry-After‘ (2 to 60 s). Show it as waiting, not failed. After 24 h of deferral a maintenance answer spends retries like any other failure, so a server stuck in maintenance still ends ‘.failed‘, revivable as before. The events ‘queue_upload_failed‘ and ‘upload_attempt_finished‘ carry ‘reason: maintenance_wait‘ while it waits.

Inspecting the queue at launch

Before you inspect, withdraw or discard queued work at launch, wait for the queue to finish loading the records an earlier launch left on disk:

```
Task {
    await sdk.awaitQueueHydrated() // the whole pass; returns at once once it has run
    for item in await sdk.getQueuedItems() where item.sessionId == abandonedSessionId {
        await sdk.removeQueuedItem(item.id)
    }
}
```

- It returns once the whole pass has finished, also when the load failed (the queue then holds what it could read). After a re-‘configure()‘ it also waits for the replaced instance’s queue to stop, so what it reports is settled.
- A setter that rebuilds the queue (‘setMaxQueueSize(_:)‘, ‘setUploadRetryCount(_:)‘, a new operations key or base URL) starts a new pass; the call waits for the queue that is current when it returns.
- On an instance that ‘configure()‘ replaced it returns at once. Call it on ‘NeurolabsSDKCore.shared‘.

Retained photos and low storage (v1.7.16)

Only pending, uploading and failed items take a ‘maxQueueSize‘ slot. Uploaded photos the queue keeps on the device (keep-photos mode, ‘setDeletePhotosOnUploadSuccess(false)‘, or photos kept until their session’s submit) never fill the queue; two process-wide settings bound them instead:

```
NLConfiguration.maxRetainedPhotos = 500     // default 500, must be > 0
NLConfiguration.retainedPhotoExpiryDays = 14 // default nil = off, must be > 0

let status = await sdk.getQueueStatus()
print("kept=\(status.retainedCount) bytes=\(status.retainedBytes)")
```

- From 80% of ‘maxRetainedPhotos‘ (rounded up) the SDK calls ‘neurolabsSDK(_:retainedPhotosHigh:limit:)‘ on ‘NeurolabsSDKDelegate‘ once per crossing (re-armed when the count drops back below) with an ‘NLRetainedPhotos‘ (‘count‘, ‘bytes‘, ‘oldestCreatedAt‘). The queue-level twin is ‘queueManager(_:retainedPhotosHigh:limit:)‘ on ‘NLQueueManagerDelegate‘. Both have default empty implementations. At the limit the SDK only logs; it never refuses a capture for retained photos.
- ‘retainedPhotoExpiryDays‘ deletes a kept uploaded photo captured at least that many days ago, files included, at the next flush or launch. Photos kept only for a submit never expire.
- A new capture is refused with ‘STORAGE_LOW‘ (retryable) when the queue’s directory has under 500 MB free. It reaches you through ‘didEncounterError‘ and the failed capture result, as ‘QUEUE_FULL‘ does, and it is a client condition (‘NLError.clientConfigurationCodes‘), never reported to the SDK’s Sentry. Since 1.7.20 the SDK’s default camera (‘openNativeCaptureUI‘) also tells the rep why it refused the photo, for ‘STORAGE_LOW‘ and ‘QUEUE_FULL‘.
- A capture taken under an injected ‘NLCameraConfiguration.frameSource‘ (a testing seam) is synthetic (‘isSynthetic‘, id prefix ‘nlsynth-‘). The queue refuses it with ‘SYNTHETIC_CAPTURE_REFUSED‘ (not retryable, a client condition, never reported to Sentry) and persists nothing; ‘onUploadRequested‘ never receives it, and its visit’s KPI payload is never submitted.

Uploads after the app is backgrounded

Since 1.7.19 the SDK hands each queued photo’s bytes to iOS: a background ‘URLSession‘ (one per queue partition) sends them to the photo’s Google Cloud Storage upload session, so the transfer continues while the app is suspended, after iOS terminates it (memory pressure, for example), and while the phone is locked. A user force-quit (swiping the app away) makes iOS cancel the app’s background transfers: nothing is lost, and the next launch resumes each photo from the offset GCS confirmed. The rest stays in your app’s process:

Step	Where it runs	Credential
‘upload/init-resumable‘, opening the GCS session	the app, when it flushes	operations Bearer
Sending the photo’s bytes	iOS (‘nsurlsessiond‘)	none: the GCS session URL itself
‘upload/complete‘, the mission submit	the app, on the wake or the next foreground	operations Bearer

The OS request carries no operations credential. iOS stores the GCS session URL with the request; it can write that one photo only and expires within 7 days. To send while the phone is locked, the SDK gives iOS a temporary copy of the photo protected ‘completeUntilFirstUserAuthentication‘ (readable once the phone has been unlocked after a restart). The queue’s own copy keeps ‘completeUnlessOpen‘. The temporary copy is deleted as soon as the transfer finishes or fails, and any copy older than 7 days is swept at launch and on every flush.

Forward the app-delegate wake to the SDK (optional, recommended):

```
func application(_ application: UIApplication,
                 handleEventsForBackgroundURLSession identifier: String,
                 completionHandler: @escaping () -> Void) {
    if NeurolabsSDKCore.handleEventsForBackgroundURLSession(identifier, completionHandler: co
mpletionHandler) {
        return
    }
    completionHandler() // not the SDK’s session: your own handling
}
```

Without it the uploads still finish in the background; the SDK learns of them the next time the app runs. ‘upload/complete‘ and the submit need the operations credential, so they run in the wake only when your app configures the SDK at launch (‘NeurolabsSDKCore.configure‘ and the operations key or auth provider in ‘application(_:didFinishLaunchingWithOptions:)‘); otherwise on the next foreground. A photo whose bytes are with iOS counts as uploading in the queue and spends none of its retry attempts. Nothing is lost when the transfer fails: the photo stays queued and resumes from its confirmed offset.

Since 1.7.20 the SDK hands a photo to iOS only while the app is in the foreground; a flush in the background (a wake, the grace period) uploads in-process, because iOS may hold a transfer created in the background for hours. With Wi-Fi-only uploads on, the transfer never uses cellular data, even when iOS finishes it with the app closed.

A flush also still holds an iOS background task (‘beginBackgroundTask‘) until it ends, so the in-app steps get iOS’s grace period (about 30 s) when the rep locks the phone or switches app.

Cordova: plugin 1.7.19 and later forward the wake itself. From plugin 1.7.20 the plugin adds the method at runtime, and only when ‘CDVAppDelegate‘ has none, so your app compiles as it does without the plugin. If your app’s own ‘AppDelegate‘ implements ‘application(_:handleEventsForBackgroundURLSession:completionHandler:)‘, yours wins: forward to ‘NeurolabsSDKCore.handleEventsForBackgroundURLSession‘ as above (in Swift, cordova-ios requires ‘override‘ on it in a ‘CDVAppDelegate‘ subclass). With an older plugin the uploads still finish in the background and complete on the next foreground.

What reaches the backend

What the operations upload carries:

- Per capture: the JPEG (the full frame, sized by ‘maxImageDimension‘ / ‘imageCompressionQuality‘), with its task and bay (‘subsection_name‘).

- Per capture, after the JPEG uploads: its capture metadata (capture contract v1, ‘PUT .../images/{imageId}/capture-metadata‘), for captures the SDK camera took: shutter time, SDK/device/OS, camera mode, photo size, an EXIF subset (no GPS, no serial numbers), device attitude, camera intrinsics and, on the AR camera, its pose. The detector’s boxes only with ‘sendDetectionsMetadata‘. Location only with ‘NLConfiguration.captureGeolocation‘ AND your app’s location permission (the SDK never asks for it); the LiDAR depth map only with ‘NLConfiguration.captureDepth‘. Both default off. It goes up from its own retrying outbox and never holds up the photo or the submit. Your app reads the same record on ‘NLNativeCaptureResult.captureMetadata‘ and ‘NLCustomCameraCapture.captureMetadata‘.
- Per session: the mission submit’s ‘client_metadata.device_dedup‘ summary for a multi-bay session (section 9).

What stays on the device:

- The capture’s detections always reach your app (‘NLCustomCameraCapture.detections‘, the capture result); ‘sendDetectionsMetadata‘ decides only whether they are kept in the queue record (‘getQueuedItems‘) and uploaded with the capture metadata.
- ‘enablePreviewCropping‘ trims only the thumbnail your app and the review show, to the area the detections cover. The uploaded photo is always the full frame, on iOS and Android.
- ‘.nlrec‘ files (‘arSessionRecording‘, experimental) are AR tooling recordings for replay and calibration. They are not a capture output and are never uploaded. A camera opened by ‘openNativeCaptureUI‘ (WebView, Cordova) records when ‘NeurolabsSDKCore.arSessionRecording‘ is set before it opens (1.7.22); set it, nil included, on every open.

7. Open Custom Camera

```
import UIKit
import NeurolabsSDK

var capture = NLCaptureConfiguration()
capture.confidenceThreshold = 0.25
capture.iouThreshold = 0.45
capture.maxCaptures = 1
capture.guidanceMode = .strict
capture.showDetectionOverlays = false
capture.showCapturedRegions = false

let config = NLCameraConfiguration.captureRange(
    capture,
    minimumCaptures: 1,
    maximumCaptures: 5,
    cameraMode: .default,
    showCapturePreview: true,
    showPreviewInStrictMode: true,
    showsCapturedCountLabel: false,
    enableValidation: true,
    showAlignmentGuidance: true,
    enableARDetections: false,
    liveQualityEnabled: true,
    liveQualityFPS: 6
)

let handlers = NLCustomCameraHandlers(
    onSave: { capture in
        // Custom post-processing hook for each accepted capture
        // return .performDefault -> SDK also enqueues upload
        // return .skipDefault -> the SDK does not queue it; the photo is
        //   yours to process, but capture.imageFilePath is only valid until
        //   this handler returns. Copy the file first (see section 10).
        return .performDefault
    },
    onSaveAll: { captures in
        // Batch hook (multi-capture flow)
        return .performDefault
    }
)

sdk.openCustomCameraUI(
    configuration: config,
    handlers: handlers,
    onClose: {
        print("Camera closed")
    }
)
```

Recommended capture payload notes:

- ‘liveQualityEnabled: true‘ is the key for pill/rotation/warning/error guidance behavior in the custom camera.
- Keep shelf capture mode with strict guidance and overlays disabled for the custom-camera guidance UI path.
- Use ‘NLCameraConfiguration.captureRange(...)‘ to express a minimum capture requirement plus a larger maximum cap.

- Use ‘showsCapturedCountLabel = false‘ to hide the count chip when you do not want ‘N captured‘ shown.
- ‘guidanceMode: .guidance‘ should only be used as a temporary fallback if a partner explicitly wants looser guidance than the parity path.

Shelf framing and Retake recommended (v1.7.28)

A still whose detections show the shelf badly framed carries one ‘NLValidationIssue‘ of kind ‘.shelfFramingPoor‘, severity ERROR. It fires when the product boxes cover under 20% of the frame, when ten or more boxes leave over 35% of the frame empty below the lowest, when the rows are rolled 8 degrees or more or converge 15 degrees or more, or, in the SDK cameras, when the live shelf yaw read 12 degrees or more just before the shutter. It needs detection in the same pass (every SDK camera, or ‘processImage‘ with ‘detect‘ and ‘validate‘); without a detector it never fires.

- It never refuses the photo: ‘isAcceptable‘, the score and the metrics are unchanged, so ‘.strict‘ keeps it too.
- The capture review reads "Retake recommended" in red for a kept photo with an ERROR issue (it read "Acceptable - minor warnings"), and the toast names the issue ("Shelf not well framed, retake").
- Tune or turn it off on ‘NLValidationOptions‘: ‘enableShelfFramingCheck‘ (true; false in ‘.minimal‘), ‘shelfFramingMinAreaShare‘ (0.20), ‘shelfFramingMaxFloorShare‘ (0.35), ‘shelfFramingMinBoxesForFloor‘ (10), ‘shelfFramingMaxRollDegrees‘ (8), ‘shelfFramingMaxConvergenceDegrees‘ (15), ‘shelfFramingMaxLiveYawDegrees‘ (12; 0 turns that leg off).
- Custom copy: the template key ‘shelfFramingPoor‘ in ‘issueLabels‘ and ‘bannerMessages‘. Android has the same rule.

Offline chip (v1.7.17)

The default and AR camera show a small chip while the upload queue sees no network: "Offline uploads queued" (or "Offline" with nothing queued), and "Waiting for Wi-Fi uploads queued" while Wi-Fi-only uploads hold photos on a cellular network. It hides while the network is unknown and when the host uploads the photos itself (‘uploadsEnabled == false‘). It is informational only and never gates the shutter.

```
config.showsOfflineIndicator = false // default true
```

On the bridge, ‘showsOfflineIndicator‘ (‘Bool?‘) on ‘NLNativeCaptureOptions‘ and ‘NLNativeCaptureConfig‘ resolves options, then config, then ‘true‘.

Capture image size (v1.7.16)

With no ‘maxImageDimension‘ / ‘imageCompressionQuality‘ set by the host, a capture is sized to a 4032 px long edge at JPEG quality 0.9. Each knob resolves on its own: the host value (‘NLCameraConfiguration‘, which ‘NLNativeCaptureOptions‘ then ‘NLNativeCaptureConfig‘ resolve into), then the org’s remote config, then the SDK default.

- The org values are ‘sdkCapture.maxImageDimension‘ and ‘sdkCapture.imageCompressionQuality‘ in ‘GET /v1/recognition-config‘. They apply only inside 1024-4032 px and 0.5-1.0; anything else is ignored (not clamped) and that knob keeps the default. They are read at capture time, so a config fetched after the camera opened applies to the next shot.

- The tightest cap wins, the aspect ratio holds and nothing is upscaled. Inside every cap an image keeps its bytes when the resolved quality is at least 0.9; a lower one (an org's 0.7, say) re-encodes it at full size. Width and height caps (‘maxImageWidth‘, ‘maxImageHeight‘) stay host-only.
- To ship the camera's JPEG untouched, set the host values: a ‘maxImageDimension‘ of 0 or less means no cap, and an ‘imageCompressionQuality‘ of 1 or more keeps an in-cap image's bytes. Host values always win over the org's remote config.

End-of-session review: finalReview (v1.7.16)

‘finalReview‘ gives the default (non-AR) camera without a multi-bay plan one review at the end of the session instead of a preview per shot:

```
var config = NLCameraConfiguration.captureRange(
    capture, minimumCaptures: 1, maximumCaptures: 10, cameraMode: .default
)
config.finalReview = true // set after init; default false

// SDK-managed / bridge: the same key, options over config, then false
var options = NLNativeCaptureOptions(allowManualFinish: true)
options.finalReview = true
sdk.openNativeCaptureUI(config: nativeConfig, sessionId: sessionId,
                        options: options)
```

- There is no per-shot preview, whatever ‘showCapturePreview‘ says. Done, or reaching the count that ends the session (‘maxCaptures‘, the auto-close target, or the required count when there is no Done button), opens **Review** with every photo.
- Tapping a photo opens it ("Photo 1 of 2"): quality readout, Crop, Delete, Previous / Next and "Back to review". A crop keeps the capture's ‘captureId‘; a delete reports ‘onCaptureDeleted‘ with source ‘.review‘.
- **Confirm** saves the photos left, in capture order with crops applied, through your ‘onSaveAll‘ / ‘onSave‘ handlers or the upload queue, and the session result is delivered as usual. **Back to camera** keeps the photos and keeps capturing.
- The close button (X) on an opened photo goes back to Review and discards nothing. The close button on Review, or the camera screen's own, closes the camera: every photo not yet confirmed is discarded (staged files and the recovery journal deleted) and the session ends as cancelled, with nothing queued. Without ‘finalReview‘ those photos would already have been queued at the shutter.
- A host ‘onClose‘ handler (‘NLCustomCameraHandlers.onClose‘) can intercept that close: the SDK discards and closes only when it returns ‘.performDefault‘.
- Nothing from the session is saved or queued before Confirm. The photos stay staged on the device (and survive an app kill) until then.
- The AR camera and multi-bay sessions ignore it and keep their own flows.

Camera-only integration

If your app presents ‘NLCustomCameraViewController‘ without creating a ‘NeurolabsSDKCore‘, register an operations credential for the detector download first. Without it (or a core) the standalone camera has no detector: capture works, but live detection, the shelf guidance cues and the overlay stay off, and the camera says "Detector unavailable. You can still capture." (reason ‘no_credentials‘), not "Detector downloading".

```
NLDetectorModels.setCredentials(apiKey: operationsKey)              // production
// NLDetectorModels.setCredentials(authProvider: provider, baseURL: stagingURL)
```

The call is idempotent (the same key again changes nothing, so a failure's retry wait still holds), a new key retries at once, and the most recent call wins, this one or a core's. A base that is not absolute https with a host is refused.

Camera analytics (1.7.18). An SDK camera follows the 'analyticsEnabled' of the 'NeurolabsSDKCore' that opened it, or, for a camera your app builds itself, of the most recently created core. A camera opened before any 'NeurolabsSDKCore' exists sends no analytics (no Mixpanel events, no event-bus 'sdk_event') for its whole lifetime, even if a core is created later. To have camera analytics, create the core first.

Camera analytics setting (1.7.19). 'NLCameraConfiguration.analyticsEnabled' (a 'Bool?', set after 'init') overrides the core for that camera. 'nil', the default, follows the core as described above. 'true' turns the camera's analytics on: an app that uses the camera without a 'NeurolabsSDKCore' sets it to opt in, and the camera then sends to the SDK's default Mixpanel tracker (the token resolves from the core's 'NLConfiguration.mixpanelToken' when a core exists, then 'NEUROLABS_MIXPANEL_TOKEN', then the Info.plist key, then the SDK default). 'false' turns the camera's analytics off, even when the core's are on. The camera's setting always wins over the core's:

- 'nil' (default): off with no core or under a core with analytics off; on, through the core's tracker, under a core with analytics on.
- 'false': off in every case.
- 'true': on in every case; through the core's tracker under a core with analytics on, otherwise through the SDK's default Mixpanel tracker.

```
var cameraConfiguration = NLCameraConfiguration(capture: NLCaptureConfiguration())
cameraConfiguration.analyticsEnabled = true   // camera-only app: opt in
```

For a signed-in rep's bearer, follow this lifecycle:

- **At sign-in,** call 'setCredentials(authProvider:)' with one provider for that rep. You can call it again on each camera open, as long as you pass that same instance. A new provider per call refreshes each time, and the new bearer counts as a new key, which clears the retry wait.
- **On account switch,** call it with a new provider.
- **On sign-out,** make the provider's 'token()' throw. There is no unregister, and a capture session asks the last provider again while the model is not on the device yet.

The download needs about 195 MB free (3.5 times the model: the archive, its unpacked copy and the compiled model are on disk at once during the install). With less it fails with 'insufficient_storage' and is retried at the next capture session. After an install the SDK deletes older model versions. If the installed model will not load, the SDK deletes it and downloads it once more; if that copy fails too, the status is 'load_failed' until the app restarts. A server answer that is not the model (a captive portal's page, a body of the wrong length) is 'download_failed', retried after a minute, not 'sha_mismatch'.

Detector download network policy (v1.7.16)

The SDK downloads the ~56 MB detector by itself (core start, a credential change, ‘setCredentials‘, a capture session opening). To keep that off cellular:

```
NLDetectorModels.prefetchPolicy = .unmeteredOnly // default .anyNetwork
// set it before creating a NeurolabsSDKCore or registering credentials
```

- With ‘.unmeteredOnly‘ those downloads run only on a network that is neither expensive (cellular, a personal hotspot) nor in Low Data Mode (Low Data Mode holds it on Wi-Fi too). A held download starts by itself when such a network appears, or at once when the policy goes back to ‘.anyNetwork‘.
- While it waits, ‘NLDetectorModels.status(for:)‘ is ‘.notDownloaded‘ and ‘NLDetectorModels.statusReason(for:)‘ is ‘waiting_for_unmetered_network‘ (‘NLDetectorPrefetchNetworkPolicy.waitingForUnmeteredNetworkReason‘), not a failure. An explicit ‘NLDetectorModels.prefetch(_:)‘ ignores the policy, and a model already on the device is used on any network.

Why the detector is unavailable (v1.7.26).
‘NLDetectorModels.statusDidChangeNotification‘ carries ‘userInfo["reason"]‘, the same value as ‘statusReason(for:)‘ for the posted status (a failure’s code, or ‘waiting_for_unmetered_network‘), absent when there is none. The camera’s chip and the AR dock name the fix for two reasons: ‘insufficient_storage‘ reads "Not enough free space. Free about 200 MB to use the product detector. You can still capture.", and ‘waiting_for_unmetered_network‘ reads "Waiting for Wi-Fi to download the product detector. You can still capture." Every other reason keeps "Detector unavailable. You can still capture." A core with no API key and no auth provider starts no download, and a download belongs to the credentials it started with: a credential you set right after the core’s init (an ‘apiKey: """‘ core, then ‘setOperationsAuthProvider‘) starts its own download instead of joining one that fails with ‘no_credentials‘. A ‘no_credentials‘ raised before any request sets no retry wait.

Quality checks without a detector (option B). While the detector is not ready (downloading, waiting for an unmetered network, no credential, or failed), the camera runs only the checks that need no detections: hold steady / tap to focus, too dark, too bright, glare, backlight, a finger over the lens, perspective skew, rotate the phone and the multi-bay overlap cues. Point at a shelf, framing, shelf-row, price-tag and product-sharpness rules stay off, and strict mode never refuses a photo for too few detections. The tick after the downloaded model swaps in mid-session turns the full set on, with no restart.

Live frames upright (v1.7.24). The camera hands its live path a buffer in sensor orientation, landscape on a portrait phone. The SDK now turns each live frame upright, as it does a photo, before the detector sees it, and maps the boxes back. Before this, live AR frames found no products: the overlay drew almost no boxes and "Point at a shelf" fired with a shelf in view. The live guidance rules, price-tag OCR and the frame stats (glare, a finger over a corner) read the frame upright too. Photos were never affected. No API change.

8. WebView Bridge Example

The same capture-range options are available through the native bridge models:

```
let config = NLNativeCaptureConfig(
    guidanceMode: .strict,
    showCloseButton: false,
    maxCaptures: 5,
    allowManualFinish: true,
    minCapturesBeforeDone: 1,
    showsCapturedCountLabel: false,
    cameraMode: .default
)

neurolabsSDK.openNativeCaptureUI(config: config, sessionId: sessionId)
```

Use ‘NLNativeCaptureOptions‘ with the same field names when the bridge payload is built from JS or other app code.

9. Multi-Bay Capture (Long Shelves)

Multi-bay capture handles shelves too long for a single photo: the user side-steps along the shelf and captures one overlapping photo per bay, the SDK guides the side-step, dedups the overlap on device, and reports merged product counts across bays in a single session result.

Enable it by setting ‘NLCameraConfiguration.multiBay‘ (nil = feature off, the default):

```
var config = NLCameraConfiguration()
config.multiBay = NLMultiBayOptions(
    minBays: 2,                   // minimum bays before the session can finish (default 1)
    maxBays: 4,                   // maximum bays per session (default 8; init clamps to 20)
    targetOverlapFraction: 0.30,  // fraction of the previous bay kept visible (default 0.30)
    minDetectionConfidence: 0.5,  // confidence floor for merge/count participation (default
0.5)
    onDeviceDedup: true,          // run overlap dedup at session finish (default true)
    generatePreview: true,        // compose the display-only stitched preview (default true)
    previewMaxDimension: 2048     // preview long-edge cap (default 2048, clamped 512...4096)
)

sdk.openCustomCameraUI(configuration: config, handlers: handlers)
```

Long continuous AR sweeps (v1.7.16). A continuous (shutter-free) sweep on the AR camera commits a bay on every auto-capture, so it may plan up to ‘NLMultiBayOptions.maxSupportedContinuousBays‘ (30). ‘init‘ still clamps to ‘maxSupportedBays‘ (8; 20 from v1.7.27); set ‘maxBays‘ after init. The capture screen clamps it to ‘minBays...30‘ for ‘captureMode == .continuous‘ on the AR camera and to ‘maxSupportedBays‘ otherwise, and logs a clamp. From v1.7.17 the bridge (‘NLNativeMultiBayConfig‘, used by the WebView bridge and Cordova) resolves ‘maxBays‘ the same way when the camera opens; in v1.7.16 it clamped to 8 on every mode.

```
var options = NLMultiBayOptions(minBays: 2)
options.maxBays = 30 // after init; past 20 honoured only for a continuous AR sweep
config.cameraMode = .ar
config.captureMode = .continuous
config.multiBay = options
```

Long guided walks (v1.7.27). A guided (shutter) walk, and any session on the default camera, may plan up to ‘NLMultiBayOptions.maxSupportedBays‘ (20; 8 before v1.7.27), set straight in ‘init‘; the default stays 8. A 10 m shelf is about 10-14 bays. The cost of a long walk does not grow with the ceiling: past 8 bays the finish-time dedup decodes each bay smaller so its working set stays at the 8-bay budget (about 39 MB), and the AR "Review shelf" pages through segments of up to 8 bays (20 7 + 7 + 6), as for a long continuous sweep. ‘previewImageData‘ is still the one panorama, so past 8 bays treat it as a thumbnail (about 205 px a bay at 20). The bridge resolves the same 20. Hosts that asked for 9-20 bays on a guided or default-camera session were clamped to 8 and now get that many: check your plan if you relied on the clamp.

```
config.multiBay = NLMultiBayOptions(minBays: 2, maxBays: 20) // guided: up to 20
```

"Review shelf" for long sweeps (v1.7.17). Past 8 bays the AR flow’s review shows the shelf as pages of consecutive bays (9 5 + 4, 30 8 + 8 + 7 + 7), each stitched when the rep turns to it, with Previous / Next and a swipe. Up to 8 bays nothing changes, and ‘NLMultiBayResult.previewImageData‘ is still the one panorama. No API change.

"On this shelf" (v1.7.17). The same review lists the walk’s facings by brand, with each brand’s share of the recognised products, and by SKU. A facing counts as recognised at a match of 0.6 or more. It needs on-device recognition (section 12). Names come from your ‘productDetailsSource‘, else from the SDK’s default names (v1.7.25, names only, so brands read "Unknown brand"); without recognition it shows the product total only. No API change.

Puzzle mode (v1.7.29). ‘NLCaptureMode.puzzle‘ (wire name ‘"puzzle"‘) is a third AR mode for coolers and displays a left-to-right walk does not fit: the rep photographs any part of the fixture, in any order. The dock points at uncovered ground and turns green at "Looks complete tap Done"; Done is allowed from ‘minBays‘ tiles on, and the plan allows up to 30 tiles. The entry chooser offers "Puzzle"; without AR it runs guided. After Done the review shows the tiles as a mosaic on the shelf plane, lists the tiles the plane could not place in a "Not placed" row and retakes a tile in place. Manual (‘.guided‘) stays the default capture mode.

On the WebView bridge the same options travel as ‘NLNativeCaptureOptions.multiBay‘ (‘NLNativeMultiBayConfig‘): every field is optional and absent fields take the ‘NLMultiBayOptions‘ defaults. The field names are camelCase and identical across iOS, Android, and Cordova.

Reading merged results:

- Native session result: ‘NLCaptureSessionResult.multiBay‘ (‘NLMultiBayResult?‘, nil unless the session ran multi-bay) carries ‘bays: [NLBaySummary]‘, ‘mergedProducts: [NLMergedProduct]‘, ‘countsByLabel: [String: Int]‘, the stitched preview (‘previewImageData‘ / ‘previewImageFileUri‘), and ‘dedupTrusted‘. From v1.7.18 it also carries ‘segments‘ (‘[NLDedupSegment]‘: ‘fromBay‘, ‘toBay‘, ‘trusted‘, ‘countsByLabel‘, one per run of linked bays) and ‘realogram‘.
- ‘NLMergedProduct.sourceDetectionIds‘ (‘[String: [String]]‘, capture id to the ids of that capture’s detections merged into the product, the representative’s own included) holds the same ids as the recognition result’s ‘detections[].detectionId‘, so you can map per-photo recognition onto unique facings. Additive.

- ‘NLMultiBayResult.layout‘ (‘NLMultiBayLayout‘: ‘.walk‘ or ‘.puzzle‘) says how the session’s photos sit on the shelf, and ‘tiles‘ (‘[NLMultiBayTile]‘: ‘captureId‘, ‘rect‘, ‘trusted‘) lists a puzzle session’s photos with where each landed on the shelf plane (‘NLPlaneRect‘, metres; nil when it could not be placed). A walk reads ‘.walk‘ with no tiles. A trusted puzzle’s ‘realogram‘ lays the tiles where the plane put them, stacked rows above each other. Additive.
- ‘countsByLabel‘ (‘counts_by_label‘ on the wire) is keyed by detector class (‘sku_single‘, ‘sku_multipack‘, price tag, promo, poster), not by product or SKU. Shelf, row and slot positions are in ‘NLMultiBayResult.realogram‘ (‘NLStitchedRealogram‘).
- **Bays line up before the dedup (v1.7.25).** iOS now seeds each pair’s registration with a translation, as Android does, so a product seen in two bays counts once. Before, a bay-to-bay step was too large for the whole-frame registration and iOS counted such a product twice. A pair that cannot be lined up breaks the link and starts a new segment. The stitched realogram lays the bays side by side. No API change.
- Use ‘NLMultiBayResult.productCount‘ for a product tally - it sums only ‘sku_single‘ + ‘sku_multipack‘; ‘countsByLabel‘ also carries price/promo/poster buckets, so summing every label over-counts products.
- **Count source (v1.7.24, AR-3, experimental).** ‘NLCameraConfiguration.multiBayCountSource‘ (bridge: ‘multiBayCountSource‘ on ‘NLNativeCaptureConfig‘ and ‘NLNativeCaptureOptions‘, ‘"aggregator"‘ or ‘"ar_tracklets"‘) picks which count ‘countsByLabel‘, ‘productCount‘ and ‘perBayCountsByLabel‘ serve. ‘.aggregator‘, the default, is the photo aggregator as before. ‘.arTracklets‘ serves the AR camera’s tracklet count: every facing counted once by its world identity across the walk. The aggregator still runs: ‘aggregatorCountsByLabel‘ / ‘aggregatorProductCount‘ always hold its counts, and ‘countSource‘ says which one served (it reads ‘.aggregator‘ when the AR count could not: not the AR camera, or tracking lost the walk). The submit summary’s ‘counts_by_label‘ stays the aggregator’s; under ‘.arTracklets‘ its ‘ar_anchor_counts‘ adds ‘count_source: "ar_tracklets"‘. The bridge summary adds ‘countSource‘ and ‘aggregatorCountsByLabel‘ only under ‘.arTracklets‘.
- Bridge / SDK-managed flows: the ‘native_capture_result‘ payload carries ‘multiBay: NLNativeMultiBayResultSummary‘ with ‘countsByLabel‘, ‘baysCount‘, ‘dedupTrusted‘, and ‘previewImageFileUri‘. Preview bytes are intentionally not sent over the bridge; the file URI is present only when a stable spilled file exists. It also carries the optional ‘realogram‘ (‘NLStitchedRealogram‘ with camelCase keys: ‘version‘, ‘coordinateSpace‘, ‘sdkVersion‘, ‘bays‘, ‘shelves[{shelfIndex, rows[{rowIndex, slots[{slotIndex, xMin, yMin, xMax, yMax, bayIndex, detectorClass, trackId, productId?, stackIndex?, score?}]}]}]‘, ‘truncated‘, ‘omittedProductCount‘) and ‘segments‘ (‘[{fromBay, toBay, trusted, countsByLabel}]‘). Each is absent when the SDK built none.
- On the bridge, ‘arAppearanceWithoutRecognition‘ (‘Bool?‘) on ‘NLNativeCaptureOptions‘ and ‘NLNativeCaptureConfig‘ resolves options, then config, then ‘true‘, into ‘NLCameraConfiguration.arAppearanceWithoutRecognition‘.

Meaning of ‘dedupTrusted‘:

- ‘true‘ - counts and ‘mergedProducts‘ are on-device deduped: each physical product instance is counted once across overlapping bays.

- ‘false‘ - an adjacent-pair alignment broke, the dedup timed out, or ‘onDeviceDedup‘ was disabled; counts are then the per-bay sum (an upper bound), not a deduped count.

Upload semantics:

- Each bay photo uploads as a normal capture through the operations queue (resumable mission flow, one image per capture).
- On ‘stopSession‘ the mission submit attaches a compact merge summary (‘{bays, counts_by_label, dedup_trusted, sdk_version, ref_homographies}‘) as structured ‘client_metadata.device_dedup‘ plus, transitionally, a ‘notes‘ duplicate for older backends. The backend stores the summary and serves ‘counts_by_label‘ only when ‘dedup_trusted‘ is true. It does not re-dedup from ‘ref_homographies‘ today.
- SDK-managed flows (‘openNativeCaptureUI‘, ‘openCustomCameraUI‘, bridge-driven capture) record that summary automatically. Only hosts presenting their own custom capture UI must pass ‘NLMultiBayResult.submitNotesJSON()‘ to ‘setMultiBaySubmitNotes(_:forSession:)‘ before ‘stopSession(_:)‘.

Config-only mode:

- There is no partner-facing in-camera mode toggle. The Single Multi-bay capture-mode chip and sheet are internal demo/debug chrome gated behind ‘NLSDKDebug.internalCameraToolsEnabled‘ (default ‘false‘; never enable it in partner builds). Partners control the mode purely via configuration: ‘multiBay‘ set = multi-bay session, ‘multiBay = nil‘ = single-bay session.
- The Neurolabs demo app enables that chrome only in local Debug builds; TestFlight/Release demo builds default to single-bay exactly like partner builds. The multi-bay feature itself is fully available in all builds via configuration - only the demo defaults changed.

10. Post-Processing Hooks

- ‘onSave‘ receives each approved capture.
- ‘onSaveAll‘ receives final batch in multi-capture mode.
- ‘onUploadRequested‘ exists but default queueing is done in ‘onSave‘ / ‘onSaveAll‘ path.

```
let handlers = NLCustomCameraHandlers(
    onSave: { capture in
        MyPostProcessor.shared.enqueue(capture)
        return .skipDefault
    }
)
```

Staged file lifetime

‘.skipDefault‘ stops the SDK queueing the photo. It does **not** hand the file over: ‘imageFilePath‘ / ‘previewImageFilePath‘ point at the camera’s staging folder, and the SDK deletes them once your handler is done with them.

| Handler | The files are valid | The SDK deletes them | |---|---|---| | ‘onSave‘ / ‘onSaveAll‘ (sync) | until the handler returns | after it returns: right away for Save All, when the camera closes for a single Save | | ‘onSaveAsync‘ / ‘onSaveAllAsync‘ | until the handler completes | after it completes, and never while it runs, even after the 30 s save watchdog unblocks the camera and the rep closes it |

- Copy or move the files you keep **before your handler returns**. In an async handler, finish reading them before you return; work you hand to another task after returning must use its own copy.
- An async handler that never completes keeps its files until a later camera's cleanup finds them older than 24 hours.
- After the watchdog the photos stay in the review, so the rep can Save again. Your late result is ignored (no double save); a retry may call your handler a second time with the same files.
- Android differs: an embedded camera's files belong to the host until it acknowledges them (PARITY-DECISIONS "Capture file lifetime").

Queue before you stop the session

If your sync ‘onSave‘ / ‘onSaveAll‘ starts ‘enqueueCapture‘ in a ‘Task‘ and returns, the camera waits only for your handler to return. With ‘autoCloseAfterCapture‘, ‘onFinish‘ arrives right after the last photo's handler returns, so a ‘stopSession(_:)‘ called at once can run before that photo is queued. Since 1.7.26 the SDK covers this: ‘stopSession(_:)‘ waits for the session's ‘enqueueCapture‘ calls already running when it is called, up to 5 s, before the visit's KPI run and the mission submit. On the bound it logs and goes on without them; it cancels nothing. An ‘enqueueCapture‘ you start after ‘stopSession(_:)‘ is not waited for.

Awaiting your own enqueues is still recommended: keep the enqueue tasks per session and await them, bounded, before the stop (the demo's ‘DemoPendingEnqueues‘), or enqueue inside ‘onSaveAsync‘ / ‘onSaveAllAsync‘, which the camera waits for (up to its 30 s save watchdog) before it finishes. An enqueue that has not started by the time you stop (a task not yet scheduled) is outside the SDK's wait, and on 1.7.25 and earlier the SDK does not wait at all.

Photo deleted (experimental)

‘onSave‘ / ‘onSaveAll‘ only ever see the final set. To hear about each photo the rep deletes, set ‘NLCameraConfiguration.onCaptureDeleted‘:

```
var config = NLCameraConfiguration()
config.onCaptureDeleted = { deletion in
    // deletion.captureId      the same id onCapture / onSave saw
    // deletion.remainingCount captures still in the session, pending + accepted
    // deletion.source         .review | .captureStrip | .bayPeek | .retake
    MyVisitLog.shared.photoDeleted(deletion.captureId, remaining: deletion.remainingCount)
}
```

- Fires exactly once per delete the rep makes, on the main actor, after the capture has left the session and its files are gone.
- ‘source‘ is ‘.review‘ for the review sheet and the final review grid, and ‘.captureStrip‘ for the per-tile delete on the capture strip. ‘.bayPeek‘ is reserved for the AR per-bay peek, which has no delete on iOS today.
- ‘source‘ is ‘.retake‘ (1.7.22) when the rep retook the photo: Retake on the photo under review, or an AR bay retake (for the photo it replaced, or a retake photo that could not take its bay). It is not a delete, but the photo is not in the result, so a host that keeps photos from ‘onCapture‘ drops it. A strict-quality reject stays silent. Handle cases added later with ‘@unknown default‘, and ignore an id you never received.

- It does not fire for the SDK's own discards: a strict-quality reject, a capture your 'onCapture' took with '.skipDefault', the camera closing, or the upload queue removing a photo (including delete-on-upload-success).
- Nil, the default, sends nothing. It is a stored property, not an 'init' parameter, so it is additive and ABI-safe. The Android SDK has the same contract under the same names.

Photo cropped (experimental)

A crop replaces the photo the rep shot. A host that keeps photos from 'onCapture' swaps its copy when it hears 'NLCameraConfiguration.onCaptureCropReplaced' (1.7.22), so it submits the crop:

```
config.onCaptureCropReplaced = { replacement in
    // replacement.replacedCaptureId  the id onCapture saw
    // replacement.capture            the cropped photo: captureId, imageData
    //                                (and imageFilePath), width, height,
    //                                detections remapped into the crop
    MyVisitStore.shared.replace(replacement.replacedCaptureId, with: replacement.capture.imag
eData)
}
```

- Fires once per applied crop, on the main actor: the per-shot review, the final review, or a bay opened from "Review shelf". A crop that did not apply sends nothing.
- The crop keeps the id today ('capture.captureId == replacedCaptureId'); key the swap on 'replacedCaptureId'. 'imageFilePath', when set, stays readable until the callback returns.
- A camera 'openNativeCaptureUI' opens reads 'NeurolabsSDKCore.onCaptureCropReplaced' instead. Android: 'NLCustomCameraView(onCaptureCropReplaced = ...)'.
- From v1.7.25 the core forgets the uncropped photo's recognition once the crop is reported: 'recognitionMatches(forCapture:)', 'unknownEmbeddings(forCapture:)' and 'priceReadings(forCapture:)' drop it. A photo that already had a recognition pass is recognised again on the crop's bytes and detections, off the main thread. A photo no pass ran for is left alone, and its pass sees the crop.

11. Delegate/Event Callbacks

```
final class SDKDelegate: NeurolabsSDKDelegate {
    func neurolabsSDK(_ sdk: NeurolabsSDKCore, didEncounterError error: NLError, sessionId: S
tring?, messageId: String?) {}
    func neurolabsSDK(_ sdk: NeurolabsSDKCore, didChangeQueueStatus status: NLQueueStatus) {}
    func neurolabsSDK(_ sdk: NeurolabsSDKCore, didUpdateLoadingProgress progress: NLLoadingPr
ogress) {}
}

sdk.delegate = SDKDelegate()
```

'neurolabsSDK(_:didProduceCaptureResult:for:sessionId:)' is optional (it has a default empty implementation). Since 1.7.13 'openNativeCaptureUI' calls it once for every capture it saves, after trying to queue it:

- 'success' is 'true' when the capture was queued, and 'message' is "Capture saved and queued for upload.".

- ‘success‘ is ‘false‘ when it could not be queued. ‘message‘ says why, and the same error also arrives on ‘didEncounterError‘.
- ‘detections‘ are the capture’s detections, whatever ‘sendDetectionsMetadata‘ says; that option decides only whether they are uploaded with the capture metadata (section 6, "What reaches the backend"). ‘captureMetadata‘ is the capture’s recorded metadata. ‘capturedRegions‘ is empty.
- It arrives after ‘didEnqueue‘ for the same capture. A capture the queue already holds is not reported again.

The whole session’s result still arrives once, at the end, on ‘onSendNativeCaptureResult‘. A camera you open yourself with ‘openCustomCameraUI‘ reports through your own handlers, not this callback.

12. On-Device Recognition (v1.7.1)

Per-account, config-gated: your operations API key resolves YOUR org’s recognition config and vector dataset - enabling an account is a backend flip, no app change. Every stage degrades gracefully; capture is never blocked by recognition.

Source consumers (SwiftPM products ‘RecognitionBootstrap‘, ‘RecognitionEngine‘, ‘RecognitionInterface‘):

```
import RecognitionBootstrap
import RecognitionInterface

let bootstrap = NLRecognitionBootstrap(configuration: .init(
    operationsApiKey: "<your operations key>",
    sdkVersion: NLConstants.sdkVersion
))
// Wrap the components at the SDK boundary (see NLRecognitionProviderBox docs):
if let c = await bootstrap.build() {
    sdk.setRecognitionProvider(NLRecognitionProviderBox(
        isReady: c.isReady,
        recognize: { pb, dets, size in
            await c.recognize(pb, dets.map {
                NLProviderDetection(label: $0.label, score: $0.score, bbox: $0.bbox, id: $0.i
d)
            }, size)
        }
    ))
}
// bootstrap.status / onStatusChange  "provider: ready (N items)" etc.
```

Short-lived session tokens. When your bearer is a session token rather than a fixed key, pass an auth provider and a stable identity instead:

```
extension MySessionTokens: NLRecognitionAuthProvider {} // token() + invalidate()

let bootstrap = NLRecognitionBootstrap(configuration: try .init(
    authProvider: sessionTokens,
    offlineIdentity: rep.id, // stable across tokens; never empty
    sdkVersion: NLConstants.sdkVersion
))
```

The bootstrap asks the provider for the bearer on every request. A 401 calls ‘invalidate()‘ and retries once; a second 401 turns recognition off. The offline record is named after a hash of ‘offlineIdentity‘, so a cold start with no token (the provider throws) still builds from the last sync. Use a new bootstrap when the rep changes.

Offline (stores with no signal). Each build that completes online saves the recognition config and the snapshot’s signed manifest for your credential. A later build that cannot reach the Operations API (offline, timeout, DNS, captive portal, 5xx) builds from it for 7 days after the last full sync (‘NLRecognitionBootstrap.offlineCacheTTL‘), if the cached embedder and every shard pass their checks. ‘bootstrap.offlineSnapshotDate‘ is then the sync time, and ‘bootstrap.statusSummary‘ reads ‘provider: ready (offline, N items, snapshot from YYYY-MM-DD)‘. A 401/403 still turns recognition off and deletes the cached build; a "disabled" config stays off. Have the rep open the app online once a day so the 7 days restart. A build that ended unavailable is retried by a later ‘build()‘ (after 60 s, or 1 h for a definitive failure) and, after a network failure, on reconnect; call ‘build()‘ again when ‘onStatusChange‘ reports ‘.ready‘. The cache lives in ‘Application Support/nl-catalog-snapshots‘ (excluded from backup).

Signed manifests (v1.7.23). Every manifest, online or from the offline cache, must carry a valid signature from a key the SDK pins; otherwise the load fails with ‘NLCatalogSnapshotLoader.LoaderError.signatureMissing‘ or ‘.signatureInvalid‘ and recognition stays off for that build. The staging key is trusted only on a Neurolabs staging operations host. An offline copy an older SDK saved is unsigned, so it is not used; the next online build replaces it.

Catalogue-mirror gate (v1.7.26). If your app keeps a local copy of the catalogue from the change feed, a new snapshot can name a product your copy does not have yet. Set the gate so the SDK keeps serving the last good build until your copy has caught up:

```
final class MirrorPosition: NLCatalogMirrorPositionProvider {
    func mirrorStatus() -> NLMirrorStatus {
        NLMirrorStatus(cursor: mirror.cursor, consecutiveFailures: mirror.failures)
    }
}
configuration.mirrorGate = NLMirrorGateConfiguration(provider: MirrorPosition())
bootstrap.onJoinDegradedChange = { reason in /* main queue */ }
```

- A snapshot is stamped with the feed position it was built at (‘catalog_position‘ in recognition-config). While your mirror’s cursor is behind it, the SDK does not ask for the new snapshot and serves the last good build from disk. It polls ‘mirrorStatus()‘ off the main thread, every 60 s and when the app returns to the foreground, only while a snapshot is held.
- The hold ends when the mirror reaches the stamp, after 7 days from the first hold since the last activation, or after 5 consecutive mirror failures (‘NLMirrorGatePolicy‘, both configurable). The two escape hatches activate the snapshot and set ‘joinDegraded‘ (‘.mirrorTimeout‘, ‘.mirrorFailures‘): resolve a recognised item missing from your copy through the server’s ‘canonical_uuids‘ lookup. With no earlier build to keep (a first install), the snapshot activates at once with ‘.noPriorSnapshot‘.

- If a snapshot is published between the config and the snapshot download, the downloaded snapshot is not the one the config announced. The SDK then fetches the config once more and gates again before activating (a snapshot ahead of your mirror is held as usual). If the two still differ, the downloaded snapshot activates with ‘joinDegraded = .snapshotMismatch‘ (1.7.28). That second config answer is handled as the first one: a 401/403 skips the build and forgets the offline cache, recognition switched off skips it as ‘recognition-config disabled‘, and an unmet ‘minSdkVersion‘ skips it; any other error (offline, 5xx, 404) falls back to ‘.snapshotMismatch‘.
- A build from the offline cache never gates. Without ‘mirrorGate‘, snapshots activate exactly as before. Android has the same types and rules.

The embedder (the pinned DINOv3-ConvNeXt-tiny export, 768-d) is downloaded by the SDK through ‘GET /v1/device-models‘ with your operations key, checked against the sha256 pinned in source, compiled on the device and cached. Do not bundle it: the SDK never scans the app bundle for a model (decision M5). When the model can’t be had (not entitled, offline, a failed download or compile) the build is skipped as ‘embedder unavailable (<reason>)‘. ‘embedderModelURL‘ is the only local override, for air-gapped or bench setups, and it must still be a 768-d ConvNeXt export.

Binary consumers: ‘RecognitionInterface‘ / ‘RecognitionEngine‘ / ‘RecognitionEngineQdrant‘ ship as xcframeworks on neurolabs-mobile-dist from v1.7.1; the bootstrap itself is source-only this release - wire the chain per the ‘NLRecognitionProviderBox‘ documentation.

Tap-to-product-card: captures expose per-detection matches via ‘sdk.recognitionMatches(forCapture:)‘; drop the SDK’s ‘NLDetectionInspectorView(photo:title:detailsSource:)‘ onto your review screen and feed it an ‘NLProductDetailsSource‘ wrapping ProductAuditKit’s ‘NLProductDetailsResolver‘ (one-liner on the source type’s docs). Offline or unresolved products fall back to a UUID + similarity row automatically. From v1.7.24 the inspector names a product only when its match scores 0.5 or more. Below that, the box keeps its class look and the detail says "Not recognised", with the closest item and its score as secondary text.

Captures your app keeps (v1.7.22): when a save handler returns ‘.skipDefault‘ and a recognition provider is set, the camera still runs recognition for the photo, without queueing it, before its save flow returns (at most 10 s), and ‘NLCapturePhoto.recognitionMatches‘ on the session result is filled. ‘openCustomCameraUI‘ and ‘openNativeCaptureUI‘ use their own core. An embedded camera (‘NLCustomCameraView‘ / ‘NLCustomCameraViewController‘ you build) uses ‘NeurolabsSDKCore.shared‘, or, while that is nil, the one live core: if your app creates more than one core, set ‘NeurolabsSDKCore.shared‘ to the core your save handlers commit through, or the camera recognises and fills nothing. To recognise a photo outside the camera (one you stored yourself), ‘await sdk.recognizeCapture(captureId:imageData:detections:)‘. It returns the matches and keeps them for ‘recognitionMatches(forCapture:)‘; nil means recognition did not run (no provider, the kill switch on, a provider not ready, or an image that does not decode), and an empty result means it ran and matched nothing.

13. AR Camera Options (v1.7.16)

All are stored properties on ‘NLCameraConfiguration‘, set after ‘init‘, with the same keys on ‘NLNativeCaptureConfig‘ and ‘NLNativeCaptureOptions‘ (resolved options, then config, then the default), except ‘productDetailsSource‘ (see below). Every default keeps the earlier look. Wire values decode leniently: "standard" is an alias of "quiet", an unknown value reads as the default and a malformed colour draws the default colour.

```
var config = NLCameraConfiguration()
config.cameraMode = .ar
config.arBoxStyle = .highlight             // .quiet (default), .identity, .highlight
config.arBoxHighlightColor = "#22C55E"     // #RRGGBB, default #22C55E
config.arBoxUnmatchedColor = "#FACC15"     // #RRGGBB, default #FACC15
config.arCoverageStyle = .tintAndOutline   // .tint (default), .outline, .tintAndOutline
config.arCoverageOutlineColor = "#FACC15"  // #RRGGBB, default #FACC15
config.arBoxPersistence = .session         // .standard (default), .session
config.arProductMarkers = .dots            // .boxes (default), .dots (experimental)
config.arLiveLabels = true                 // experimental, default false
config.arLiveCount = true                  // experimental, default false
config.arLiveKPIRings = true               // experimental, default false (AR-4)
config.productDetailsSource = detailsSource // names for arLiveLabels
```

- ‘arBoxStyle = .highlight‘ draws every world-locked box as a solid outline in ‘arBoxHighlightColor‘, with no fill. ‘NLARBoxStyle‘ is non-frozen: an exhaustive ‘switch‘ over it needs ‘@unknown default‘.
- With ‘arLiveLabels‘ on and a ready recognition provider, a highlight box that is stable, has been looked at, and has no product name latched on it (no match, or every look below 0.6) is drawn in ‘arBoxUnmatchedColor‘ instead. A box not looked at yet stays ‘arBoxHighlightColor‘. With the labels off, every box keeps ‘arBoxHighlightColor‘.
- ‘arCoverageStyle‘ shows the captured shelf as a tint, an outline on the fitted shelf plane in ‘arCoverageOutlineColor‘ around the captured bays, or both. With no plane fit yet every style shows the tint.
- ‘arBoxPersistence = .session‘ keeps every box that settled for the rest of the session as a dimmed outline (at most 300). ‘.standard‘ fades a box once tracking drops it. Presentation only: counts and coverage are unchanged.
- ‘arProductMarkers = .dots‘ (experimental, v1.7.29) draws one small dot per product at its centre instead of a box. A product still settling is a faint small white dot, a settled one a larger white dot, one inside a photo already taken turns orange with a dark hairline, and one the live labels looked at but could not name (‘arLiveLabels‘) is a hollow ring; each state differs in size or shape, not only in colour. At most the 64 dots nearest the screen centre are drawn. Dots replace the boxes (‘arBoxStyle‘ has no effect with dots); coverage, counts and the live labels are unchanged. New dots and dots turning orange pop in; with Reduce Motion they appear at once, and when the phone runs hot (thermal state serious or critical) the pop is dropped. Wire key ‘arProductMarkers‘, values "boxes" and "dots". ‘NLARProductMarkerStyle‘ is non-frozen: an exhaustive ‘switch‘ over it needs ‘@unknown default‘.

- ‘arLiveLabels‘ draws the top-1 product name on each stable box, only at a match score of 0.6 or more. It needs a ready recognition provider (‘setRecognitionProvider(_:)‘, section 12), and it stops while the organisation’s on-device recognition switch is off. It never runs on the AR appearance fallback (the colour histogram or the appearance-only embedder), which cannot name a product. Names come from ‘productDetailsSource‘ (an ‘NLProductDetailsSource‘, the seam ‘NLDetectionInspectorView‘ takes), else from the SDK’s default names (v1.7.25, below). Without a provider, or without any name, there are no labels.
- **Default names (v1.7.25, experimental).** When the camera has no ‘productDetailsSource‘, the SDK names the matched catalog UUIDs itself, for the live labels and "On this shelf":

- It asks ‘GET /v1/catalog-items?canonical_uuids=‘ on the operations host, at most 100 UUIDs a request, with the SDK’s own operations credential (the API key, or your ‘setOperationsAuthProvider‘ bearer). The request never leaves the operations origin; a redirect to another host drops the credential. - It keeps names only (no brand, image URL or barcode) in Application Support, excluded from backup: one file per organisation credential, named by a salted hash of the operations host and the API key (or, with an auth provider, the identity you pass to ‘setStoreContextIdentity‘). With an auth provider and no identity, names stay in memory for that credential only. - A name is used for 7 days, then asked again; offline, the last known name is used for up to 30 days. No name, no label text: the SDK never guesses. At most 5,000 names a credential, least recently used first out. - "On this shelf" gets names only from it; brands read "Unknown brand" unless your own source supplies them. - Your ‘productDetailsSource‘ always wins. To turn the default off (no requests, nothing kept), set ‘sdk.defaultProductNamesEnabled = false‘ before opening the camera: it is read when the camera opens. Android: ‘NeurolabsSDKCore.defaultProductNamesEnabled‘.

- ‘productDetailsSource‘ on ‘NLCameraConfiguration‘ reaches a camera you open with ‘openCustomCameraUI‘ or present as ‘NLCustomCameraViewController(configuration:)‘ (the camera-only path, section 7). ‘openNativeCaptureUI‘ builds its own configuration, so from v1.7.17 a bridge host sets ‘NeurolabsSDKCore.productDetailsSource‘ before opening the camera instead; a source on the configuration wins. It cannot travel in the bridge’s JSON. Before v1.7.17 ‘arLiveLabels‘ showed no names through ‘openNativeCaptureUI‘.
- ‘arLiveCount‘ shows a "~ N products" chip: the distinct stable boxes seen this session. It is an estimate; the review keeps the exact deduplicated count.
- ‘arLiveKPIRings‘ (v1.7.24, AR-4) shows three small rings on the AR multi-bay camera: the SKU facings counted so far (filled by how many have settled), the gaps seen in the captured bays (filled by their share), and price tags per facing as a percentage. They are estimates from what the overlay already tracks, refreshed about twice a second; no KPI engine runs while capturing. They hide while the guidance dock is amber or red and during a review, and VoiceOver reads them as one element.
- With recognition off, there are no product names or unmatched colour, and a box is told apart from its neighbours by its position, not by which product it is: two identical facings that touch can be harder to keep apart. From v1.7.18 ‘arAppearanceWithoutRecognition‘ (on by default) adds appearance to that: the same ConvNeXt model, downloaded for appearance only for an org Operations grants it to, or else a colour histogram. The default camera never downloads it.

Long walks (v1.7.24). A settled facing that leaves the camera's view is archived: it keeps its id and count, no longer takes live track budget, and comes back under its old id when the camera returns. Price tags, promos and posters get a smaller share of the budget (64 of 256). So long and dense walks no longer run out of budget, and the last bays get boxes. A facing seen square on turns solid after one press. Under '.v1' the multi-bay AR count ('ar_anchor_counts') counts archived facings too. No API change.

Detector gate and review (v1.7.24). The AR multi-bay walk holds auto and manual capture under "Getting ready..." until the live detector has run with its model loaded, for 45 s at most. Past that, or at once when the model cannot arrive, capture goes ahead and a chip says "Detector unavailable. You can still capture." When recognition ran and nothing matched, the review says "No catalogue matches 0 of N facings recognised". A remembered Guided / Auto pick now applies only under the host default it was made under, so a continuous preset runs continuous. A pick stored before 1.7.24 carries no default, so each install sees the chooser once more. No API change.

12 MP AR stills. The AR shutter (single, multi-bay, guided and continuous) queues ARKit's full-resolution still ('ARSession.captureHighResolutionFrame', 12 MP on current iPhones) instead of the ~1920x1440 video frame, while tracking keeps running. It goes through the same image sizing (4032 px / JPEG 0.9 by default, section 7). No API change.

- It falls back to the video frame when there is no running session ('no_session'), on an ARKit error ('capture_error'), with no still within 1.5 s ('timeout'), for a still in the other orientation ('orientation_mismatch'), or for a still no larger than the video frame ('not_larger'). The reason goes to the device log only.
- Until ARKit returns the still, the shutter shows a spinner and is disabled, so a second tap does nothing and nothing is queued.

14. Notes

Store context for visit KPIs (since 1.7.21)

A visit's KPIs need the outlet's store context (targets, assortment, planograms and the org's metric definitions). The SDK fetches, verifies and caches it per credential for the org's offline TTL (default 7 days). Outlet ids are Operations outlet UUIDs.

```
let result = await sdk.prefetchStoreContext(outletIds: routeOutletIds)  // never throws; resu
lt.refreshFailure is a code
let status = await sdk.storeContextStatus(outletId: outletId)           // cache only, no net
work
sdk.setStoreContextIdentity(currentUser.id)  // only with setOperationsAuthProvider; hashed,
never stored or sent
await sdk.clearStoreContextCache()            // sign-out
```

'status.state' is '.ready', '.expired' or '.missing' (switch with '@unknown default'), and 'status.reason' is a stable code ('not_assigned' means the outlet id is not your org's; treat an unknown code as missing). 'device_kpis' are submitted only for a context that is ready and 'isSignatureVerified'; the SDK pins the production key 'store-context-v1' and requires signatures. Staging-backed builds verify with the staging key 'store-context-v2', trusted only on a Neurolabs staging operations host; a production base only ever trusts v1. The full code list is in the README ("Store context (visit KPIs)").

Visit KPIs (since 1.7.21)

After an SDK-managed session finishes with at least one capture, the SDK computes the visit's KPIs on the device and calls 'neurolabsSDK(_:didComputeVisitKPIs:sessionId:)' (default no-op, main actor). On by default; 'setVisitKPIsEnabled(false)' turns it off.

With your own capture UI ('NLCustomCameraView', or photos you queue with 'enqueueCapture'), 'stopSession(_:)' computes them once instead (1.7.23), from the photos queued for that session, under the same rules. Set the context on the session you stop. Since 1.7.24 what the run needs is recorded on disk, so a visit that spans an app kill counts every photo, with its recognition matches, once you call 'stopSession(_:)' after the relaunch. A session the SDK submits on its own after a relaunch, with no 'stopSession', goes without 'device_kpis'.

```
sdk.setVisitKPIContext(NLVisitKPIContext(outletId: outletId, sectionsByCaptureId: [:], sectio
nsByBayIndex: [:]),
                       forSession: sessionId)          // before the session finishes; refres
hes the store context
let result = try await sdk.computeVisitKPIs(for: sessionResult, context: context)  // custom
capture UI; read-only
let preview = try await sdk.computeVisitKPIs(for: sessionResult, context: context,   // a sto
re context you supply:
                                             storeContext: NLVisitKPIHostStoreContext(storePa
rtJSON: storePart, orgPartJSON: orgPart))
```

Price-tag pairing (1.7.24). A visit's 'tag_paired' (K16) now pairs each 'price' or 'promo' tag with at most one product by shelf geometry, measured in the product's own size: the tag sits on the shelf edge just under the product. Before, it used a fixed pixel reach and almost never paired on a full-resolution photo, so K16 read about 0 % and most positions got an 'add_tag' action. Android and zia-operations' reference core use the same rule. The recognition payload's 'priceMatches' are unchanged. From 1.7.25 'priceReadings(forCapture:)[].pairedDetectionId' uses the same pairing, so a host's price and K16 name the same product. A peg tag, a tag under a gap and a tag outside every band pair nothing.

'computeVisitKPIs(for:context:storeContext:)' (1.7.24) runs the same engine on a store part ('nl.store_context' v1) and org part ('nl.store_context_org' v1) you pass as JSON, for previews, demos and tests. Their shape is checked, never a signature, so the result is never submitted; a part that fails a check throws 'VISIT_KPIS_STORE_CONTEXT_INVALID'.

Visit KPIs by session id (1.7.25)

'computeVisitKPIs(forSession:context:storeContext:)' runs the same pull when you have only the session id, as a Cordova host does:

```
let preview = try await sdk.computeVisitKPIs(forSession: sessionId, context: nil,   // nil: t
he context you set for it
                                             storeContext: NLVisitKPIHostStoreContext(storePa
rtJSON: storePart, orgPartJSON: orgPart))
```

- The session must have been finished by a camera the SDK runs ('openNativeCaptureUI', the Cordova plugin) in this process, with at least one photo; a cancelled session counts. The SDK holds the newest 8 such sessions in memory: detections, sizes, times, recognition matches, the multi-bay result and its dedup setting. It keeps no pixels and writes nothing to disk.

- For a session you queued yourself (‘enqueueCapture‘), it reads the photos queued for it, until its ‘stopSession‘ run has taken them.
- The result is the one ‘computeVisitKPIs(for:context:storeContext:)‘ gives for the same session, and is never submitted. The one difference: it uses the camera’s ‘onDeviceDedup‘, where the captures-based call takes it as on, so a walk captured with on-device dedup off gives a different result.
- A nil ‘context‘ uses the one you set with ‘setVisitKPIContext(_:forSession:)‘, even after ‘stopSession‘.
- An id the SDK holds nothing for (another process, no photo, or 8 newer sessions since) throws ‘VISIT_KPIS_SESSION_UNKNOWN‘. ‘VISIT_KPIS_DISABLED‘ and ‘VISIT_KPIS_STORE_CONTEXT_INVALID‘ come first.

‘NLVisitKPIResult‘ carries the canonical ‘nl.visit_kpis‘ v1 JSON (‘payloadJSON‘), ‘isProvisional‘ (always true on the device: label it "Provisional"), ‘storeContext‘ (‘.used(offline:)‘, ‘.missing‘, ‘.expired‘), ‘perfectStoreScoreBp‘, ‘actionCount‘ and ‘willSubmit‘. Without a store context, show gaps and price-tag presence only, as "Needs store targets". The payload rides the mission submit as ‘client_metadata.device_kpis‘ only when the store context was ready and signature-verified and the payload passes the backend’s checks; a ‘400 invalid_device_kpis‘ makes the SDK resubmit once without it. Details are in the README ("Visit KPIs").

Theming the SDK screens

The SDK’s screens read ‘NLTheme.current‘, and its default is the ZIA Lens v3 design system (orange action colour, ink text, 8/16 pt radii); the SDK ships no fonts, so Geist, Geist Mono and Plus Jakarta Sans are used only when your app registers them, and the system faces stand in otherwise. To keep the look from before Lens v3, set ‘NLTheme.current = .neutral‘ once at setup, or assign your own ‘NLTheme‘. For a partial override start from a preset, ‘var t = NLTheme.neutral; t.tint = myColor; NLTheme.current = t‘, because ‘NLTheme(tint: myColor)‘ fills every other token with the Lens defaults. ‘success‘, ‘info‘, ‘warning‘ and ‘danger‘ are a separate signal palette (green, blue, orange, red) for detection boxes and status tones, and ‘NLTypeRole‘ with ‘theme.font(.body)‘ gives the Lens type ramp with Dynamic Type. Text on the action colour follows ‘onTint‘ (automatic: ink, or white when ink would not reach 4.5:1 against your ‘tint‘). Your own ‘textFont‘ and ‘monoFont‘ closures serve the fixed-size calls ‘textFont(size:weight:)‘ and ‘monoFont(size:weight:)‘; the Dynamic Type calls (‘theme.font(...)‘ and the ‘relativeTo:‘ overloads) never call them: they use the brand faces when registered, or the system faces scaled with Dynamic Type when you set ‘usesBrandFaces = false‘. The AR capture camera follows the theme too: by default it draws the Lens v3 chrome (the ‘surfaceCamera‘ plate, the orange CTA with ink text, orange progress and in-position, Lucide glyphs); under ‘NLTheme.neutral‘ it restores the previous camera (the ‘#101215‘ plate, cyan progress, green ready, the white capsule Finish / Done and the previous radii) with the Lucide glyphs and the system faces. The standard (non-AR) camera’s guidance keeps its own colours either way.

Telemetry and privacy (Sentry)

The SDK reports errors to the Neurolabs Sentry project under this policy, the same on iOS, Android and Cordova:

- **Only the SDK's own errors and crashes.** The SDK reports the errors it captures itself, and crashes whose crashed thread runs SDK code. A crash in your own code, an app hang, a watchdog termination or a failed request of your app is never sent.
- **Your app is never instrumented.** No swizzling, and no network, app-hang, watchdog, breadcrumb or session tracking. The SDK's errors go through a private Sentry client, so your own Sentry setup is untouched. The one process-global path is 'NLSentryCrashReporter.bootstrap(...)', which installs a crash handler only when your app (or the Cordova plugin) calls it; it applies the same filter. It never replaces a Sentry client your app already started: if one is running, 'bootstrap' starts nothing and the SDK reports its own errors through its private client only. If your app starts Sentry *after* 'bootstrap', yours replaces the SDK's (Sentry's global start is last-writer-wins), and the SDK's own errors still go through its private client.
- **No credentials or personal data.** 'sendDefaultPii' is off. Messages are scrubbed before they are sent: a URL keeps only its scheme, host and path, so a signed URL's credentials never leave the device. Since 1.7.20 an error that came from a server response carries only the HTTP status and the server's 'error_code', never the response body.
- **Opt out** with 'sentryEnabled: false' (or 'analyticsEnabled: false') on 'NLConfiguration'. Nothing is sent from an XCTest process.

Other notes

- Per-session routing is supported via native capture config task UUID overrides.
- Ensure 'NSCameraUsageDescription' is present in app 'Info.plist'.